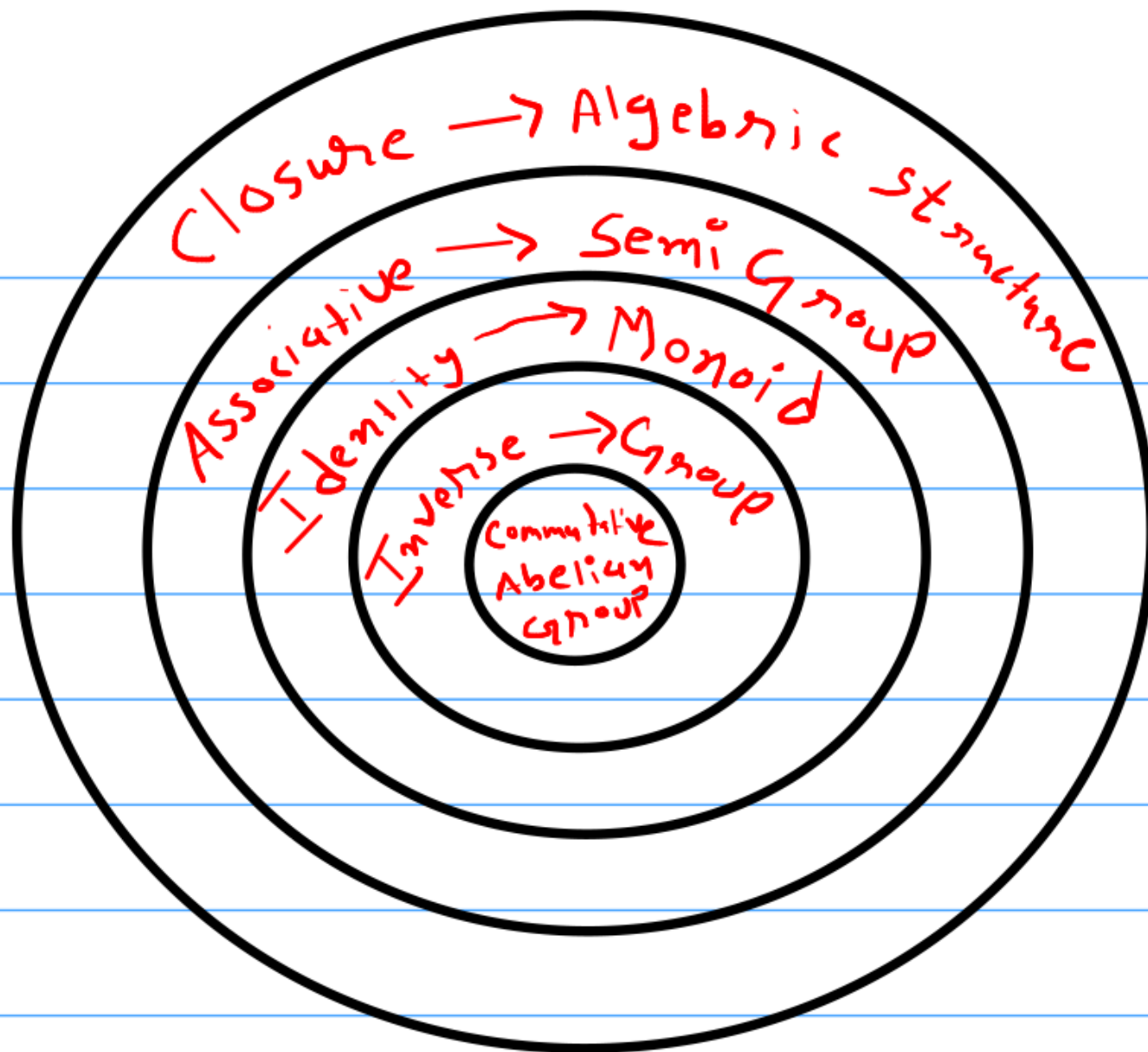


Group Theory

Introduction

- Set theory : Study of set operations.
- Relations : Study of pairs generated by product of two sets.
- Group Theory : Study of set with respect to some operation.



Algebraic Structure

- A non empty set S is called an algebraic structure w.r.t binary operation $(*)$ if it follows following axioms:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.

Let $S = \{-1, 1\}$

is $(S, *)$ algebraic structure?

$$-1 \times -1 = 1$$

$$-1 \times 1 = -1$$

$$1 \times 1 = 1$$

$$1 \times -1 = -1$$

$\in S$

So, YES

Commonly used sets

- **Natural Numbers (N)**, (also called positive integers, counting numbers, or natural numbers); They are the numbers $\{1, 2, 3, 4, 5, \dots\}$
- **Integers (Z)**. This is the set of all whole numbers plus all the negatives (or opposites) of the natural numbers, i.e., $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- **Rational numbers (Q)**. This is all the fractions where the top and bottom numbers are integers; e.g., $1/2, 3/4, 7/2, -4/3, 4/1$ [Note: The denominator cannot be 0, but the numerator can be].
- **Real numbers (R)**, (also called measuring numbers or measurement numbers). This includes all numbers that can be written as a decimal. This includes fractions written in decimal form e.g., 0.5, 0.75 2.35, -0.073 , 0.3333, or 2.142857. It also includes all the irrational numbers such as π , $\sqrt{2}$ etc. Every real number corresponds to a point on the number line.

Algebraic Structure Examples

	Algebraic System
$\mathbb{N}, +$	
$\mathbb{N}, -$	
$\mathbb{N}, *$	
$\mathbb{N}, /$	
$\mathbb{Z}, +$	
$\mathbb{Z}, -$	
$\mathbb{Z}, *$	
$\mathbb{Z}, /$	
$\mathbb{R}, +$	
$\mathbb{R}, -$	
$\mathbb{R}, *$	
$\mathbb{R}, /$	
$\mathbb{E}, +$	
$\mathbb{E}, *$	
$\mathbb{O}, +$	
$\mathbb{O}, *$	
$\mathbb{M}, +$	
$\mathbb{M}, *$	

	Algebraic System
$N, +$	Yes
$N, -$	No
$N, *$	Yes
$N, /$	No
$Z, +$	Yes
$Z, -$	Yes
$Z, *$	Yes
$Z, /$	No
$R, +$	Yes
$R, -$	Yes
$R, *$	Yes
$R, /$	No
$E, +$	Yes
$E, *$	Yes
$O, +$	No
$O, *$	Yes
$M, +$	Yes
$M, *$	Yes

Semigroup

- A non-empty set S , $(S, *)$ is called a semigroup if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c \text{ belongs to } S$.

	Algebraic System	Semigroup
$\mathbb{N}, +$	Yes	
$\mathbb{N}, -$	No	
$\mathbb{N}, *$	Yes	
$\mathbb{N}, /$	No	
$\mathbb{Z}, +$	Yes	
$\mathbb{Z}, -$	Yes	
$\mathbb{Z}, *$	Yes	
$\mathbb{Z}, /$	No	
$\mathbb{R}, +$	Yes	
$\mathbb{R}, -$	Yes	
$\mathbb{R}, *$	Yes	
$\mathbb{R}, /$	No	
$\mathbb{E}, +$	Yes	
$\mathbb{E}, *$	Yes	
$\mathbb{O}, +$	No	
$\mathbb{O}, *$	Yes	
$\mathbb{M}, +$	Yes	
$\mathbb{M}, *$	Yes	

	Algebraic System	Semigroup
$\mathbb{N}, +$	Yes	Yes
$\mathbb{N}, -$	No	No
$\mathbb{N}, *$	Yes	Yes
$\mathbb{N}, /$	No	No
$\mathbb{Z}, +$	Yes	Yes
$\mathbb{Z}, -$	Yes	No
$\mathbb{Z}, *$	Yes	Yes
$\mathbb{Z}, /$	No	No
$\mathbb{R}, +$	Yes	Yes
$\mathbb{R}, -$	Yes	No
$\mathbb{R}, *$	Yes	Yes
$\mathbb{R}, /$	No	No
$\mathbb{E}, +$	Yes	Yes
$\mathbb{E}, *$	Yes	Yes
$\mathbb{O}, +$	No	No
$\mathbb{O}, *$	Yes	Yes
$\mathbb{M}, +$	Yes	Yes
$\mathbb{M}, *$	Yes	Yes

Monoid

- A non-empty set S , $(S, *)$ is called a monoid if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \ \forall \ a, b, c$ belongs to S .
 - **Identity Element:** There exists $e \in S$ such that $a*e = e*a = a \ \forall \ a \in S$

	Algebraic System	Semigroup	Monoid
$\mathbb{N}, +$	Yes	Yes	
$\mathbb{N}, -$	No	No	
$\mathbb{N}, *$	Yes	Yes	
$\mathbb{N}, /$	No	No	
$\mathbb{Z}, +$	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	
$\mathbb{Z}, *$	Yes	Yes	
$\mathbb{Z}, /$	No	No	
$\mathbb{R}, +$	Yes	Yes	
$\mathbb{R}, -$	Yes	No	
$\mathbb{R}, *$	Yes	Yes	
$\mathbb{R}, /$	No	No	
$\mathbb{E}, +$	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	
$\mathbb{O}, +$	No	No	
$\mathbb{O}, *$	Yes	Yes	
$\mathbb{M}, +$	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	

	Algebraic System	Semigroup	Monoid
$\mathbb{N}, +$	Yes	Yes	No
$\mathbb{N}, -$	No	No	No
$\mathbb{N}, *$	Yes	Yes	Yes
$\mathbb{N}, /$	No	No	No
$\mathbb{Z}, +$	Yes	Yes	Yes
$\mathbb{Z}, -$	Yes	No	No
$\mathbb{Z}, *$	Yes	Yes	Yes
$\mathbb{Z}, /$	No	No	No
$\mathbb{R}, +$	Yes	Yes	Yes
$\mathbb{R}, -$	Yes	No	No
$\mathbb{R}, *$	Yes	Yes	Yes
$\mathbb{R}, /$	No	No	No
$\mathbb{E}, +$	Yes	Yes	Yes
$\mathbb{E}, *$	Yes	Yes	No
$\mathbb{O}, +$	No	No	No
$\mathbb{O}, *$	Yes	Yes	Yes
$\mathbb{M}, +$	Yes	Yes	Yes
$\mathbb{M}, *$	Yes	Yes	Yes

Group

- A non-empty set G , $(G, *)$ is called a group if it follows the following axiom:
 - **Closure:** $(a * b)$ belongs to G for all $a, b \in G$.
 - **Associativity:** $a * (b * c) = (a * b) * c \quad \forall a, b, c$ belongs to G .
 - **Identity Element:** There exists $e \in G$ such that $a * e = e * a = a \quad \forall a \in G$
 - **Inverses:** $\forall a \in G$ there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

	Algebraic System	Semigroup	Monoid	Group
$\mathbb{N}, +$	Yes	Yes	No	
$\mathbb{N}, -$	No	No	No	
$\mathbb{N}, *$	Yes	Yes	Yes	
$\mathbb{N}, /$	No	No	No	
$\mathbb{Z}, +$	Yes	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	No	
$\mathbb{Z}, *$	Yes	Yes	Yes	
$\mathbb{Z}, /$	No	No	No	
$\mathbb{R}, +$	Yes	Yes	Yes	
$\mathbb{R}, -$	Yes	No	No	
$\mathbb{R}, *$	Yes	Yes	Yes	
$\mathbb{R}, /$	No	No	No	
$\mathbb{E}, +$	Yes	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	No	
$\mathbb{O}, +$	No	No	No	
$\mathbb{O}, *$	Yes	Yes	Yes	
$\mathbb{M}, +$	Yes	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	Yes	

Abelian Group or Commutative group

- A non-empty set S , $(S, *)$ is called a Abelian group if it follows the following axiom:
 - **Closure:** $(a*b)$ belongs to S for all $a, b \in S$.
 - **Associativity:** $a*(b*c) = (a*b)*c \quad \forall a, b, c$ belongs to S .
 - **Identity Element:** There exists $e \in S$ such that $a*e = e*a = a \quad \forall a \in S$
 - **Inverses:** $\forall a \in S$ there exists $a^{-1} \in S$ such that $a*a^{-1} = a^{-1}*a = e$
 - **Commutative:** $a*b = b*a$ for all $a, b \in S$

	Algebraic System	Semigroup	Monoid	Group	Abelian Group
$\mathbb{N}, +$	Yes	Yes	No	No	
$\mathbb{N}, -$	No	No	No	No	
$\mathbb{N}, *$	Yes	Yes	Yes	No	
$\mathbb{N}, /$	No	No	No	No	
$\mathbb{Z}, +$	Yes	Yes	Yes	Yes	
$\mathbb{Z}, -$	Yes	No	No	No	
$\mathbb{Z}, *$	Yes	Yes	Yes	No	
$\mathbb{Z}, /$	No	No	No	No	
$\mathbb{R}, +$	Yes	Yes	Yes	Yes	
$\mathbb{R}, -$	Yes	No	No	No	
$\mathbb{R}, *$	Yes	Yes	Yes	No	
$\mathbb{R}, /$	No	No	No	No	
$\mathbb{E}, +$	Yes	Yes	Yes	Yes	
$\mathbb{E}, *$	Yes	Yes	No	No	
$\mathbb{O}, +$	No	No	No	No	
$\mathbb{O}, *$	Yes	Yes	Yes	No	
$\mathbb{M}, +$	Yes	Yes	Yes	Yes	
$\mathbb{M}, *$	Yes	Yes	Yes	No	

	Algebraic System	Semigroup	Monoid	Group	Abelian Group
$\mathbb{N}, +$	Yes	Yes	No	No	No
$\mathbb{N}, -$	No	No	No	No	No
$\mathbb{N}, *$	Yes	Yes	Yes	No	No
$\mathbb{N}, /$	No	No	No	No	No
$\mathbb{Z}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{Z}, -$	Yes	No	No	No	No
$\mathbb{Z}, *$	Yes	Yes	Yes	No	No
$\mathbb{Z}, /$	No	No	No	No	No
$\mathbb{R}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{R}, -$	Yes	No	No	No	No
$\mathbb{R}, *$	Yes	Yes	Yes	No	No
$\mathbb{R}, /$	No	No	No	No	No
$\mathbb{E}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{E}, *$	Yes	Yes	No	No	No
$\mathbb{O}, +$	No	No	No	No	No
$\mathbb{O}, *$	Yes	Yes	Yes	No	No
$\mathbb{M}, +$	Yes	Yes	Yes	Yes	Yes
$\mathbb{M}, *$	Yes	Yes	Yes	No	No

- $(\mathbb{N}, *)$ where $\mathbb{N}$ is natural number. $*$ refers to the power operation.
 - It is semi group
 - It is not semi group
 - It is monoid but not group
 - It is Group

- $(N, *)$ where N is natural number. $*$ refers to the power operation.
 - It is semi group
 - **It is not semi group**
 - It is monoid but not group
 - It is Group

- $(Z, *)$ where Z is natural number. $*$ refers to the Max operation.
 - Algebraic structure ? Semigroup ? Group ?
- $(Z, *)$ where Z is natural number. $*$ refers to the Min operation.
 - Algebraic structure ? Semigroup ? Group ?

- $(Z, *)$ where Z is natural number. $*$ refers to the Max operation.
 - Algebraic structure ? **Semigroup** ? Group ?
- $(Z, *)$ where Z is natural number. $*$ refers to the Min operation.
 - Algebraic structure ? **Semigroup** ? Group ?

~ Let $Q^+ =$ Set of Positive Rational No.
 $a * b = \frac{a \cdot b}{3}$, $(Q^+, *)$ is abelian Group.

a) $e = 3$

b) $a^{-1} = \frac{9}{a}$

3) $(\frac{2}{3})^{-1} = 6$

4) $3^+ = 3$

- Which of the following is not a group?

a) $\{0, \pm 2, \pm 4, \pm 6, \dots, \infty\}, +$

b) $\{0, \pm k, \pm 2k, \pm 3k, \dots, \infty\}, +$ (k is Natural No.)

c) $\{2^n \mid n \text{ is integer}\}, *$

d) $\{\text{Set of all Complex no}, *\}$

Subgroup

- A **subgroup** of a group G is a subset of G that forms a group with the same law of composition. For example, the even numbers form a subgroup of the group of integers with group law of addition.
- The group of integers equipped with addition is a subgroup of the real numbers equipped with addition; i.e. $(\mathbb{Z}, +) \subset (\mathbb{R}, +)$

Addition modulo Operator

* Addition Modulo ($+_m$)

$$a +_m b = \begin{cases} a + b & \text{if } (a + b < m) \\ (a + b) - m & \text{if } (a + b > m) \end{cases}$$

$$S = \{0, 1, 2, 3\}, \text{ OP} = +_4$$

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$S_0, (S, +_4)$ is Group.

Multiplication modulo Operator

Similarly, Multiplication Modulo

$$a *_m b = \begin{cases} a * b & \text{if } (a * b < m) \\ (a * b) \div m & \text{if } (a * b \geq m) \end{cases}$$

Ex. $S = \{1, 2, 3, 4\}$, $*_5$

Q. Which of the following is Group?

$$\{0, 1, 2, 3\}, *_4$$

$$\{1, 2, 3\}, *_4$$

$$\{1, 2, 3, 4, 5, 6\}, *_7$$

$$\{2, 3, 4, 5, 6\}, *_7$$

$$\{1, 3, 5, 7\}, *_8$$

$$\{1, 2, 4, 7, 8, 11, 13, 14\}, *_{15}$$

Order of an Element

* Let $(G, *)$ be a group. Order is defined for $\forall a \in G$,
$$o(a) = n$$

~ n is Smallest Positive integer which satisfies the eq. $a^n = e$.

Let, $G = \{0, 1, 2, 3\}, +_4$

Identity(e) for this group is 0.

Identity(e) for this group is 0.

$$0^1 = 0$$

$$0^2 = 0 +_4 0 = 0$$

$$0^3 = 0 +_4 0 +_4 0 = 0$$

$$0^4 = 0 +_4 0 +_4 0 +_4 0 = 0$$

$$S_0, \boxed{O(0) = 1}$$

$$1^1 = 1$$

$$1^2 = 1 +_4 1 = 2$$

$$1^3 = 1 +_4 1 +_4 1 = 3$$

$$1^4 = 1 +_4 1 +_4 1 +_4 1 = 0$$

$$\boxed{O(1) = 4}$$

$$2^1 = 2$$

$$2^2 = 2 +_4 2 = 0$$

$$2^3 = 2 +_4 2 +_4 2 = 2$$

$$\boxed{O(2) = 2}$$

$$3^1 = 3$$

$$3^2 = 3 +_4 3 = 2$$

$$3^3 = 3 +_4 3 +_4 3 = 1$$

$$3^4 = 3 +_4 3 +_4 3 +_4 3 = 0$$

$$\boxed{O(3) = 4}$$

Cyclic Group and Generating Element

* On a group $(G, *)$, an element is called generator if $a \in G$, and $\forall g \in G$ can be generated by power of a .

$$G = \{0, 1, 2, 3\}, +_4$$

$$1^1 = 1$$

$$1^2 = 2$$

$$1^3 = 3$$

$$1^4 = 0$$

$$1^5 = 1$$

} 1 can generate all elements of G .

* Cyclic Group: A Group Which has at least one generating element.

$G = \{0, 1, 2, 3\}, +_4$ is Cyclic Group.

Let $G = \{1, 3, 5, 7\} *_8$

We note that above Group is not Cyclic.

Coset

~ Let G be Multiplicative Group.

H Subgroup.

Let ' a ' be any element of G .

then.

$H \cdot a = \{h \cdot a, h \in H\}$ - Right Coset

$a \cdot H = \{a \cdot h, h \in H\}$ - Left Coset

- In Case of additive Group

$H + a = \{h + a, h \in H\}$ Right Coset

$a + H = \{a + h, h \in H\}$ Left Coset

~ Let G be Set of Real Numbers

H is Set of Integers $= \{0, \pm 1, \pm 2, \dots\}$

Let $\frac{1}{2} = a \in G$

Right Coset $= H + a = H + \frac{1}{2} = \{h + \frac{1}{2}, h \in H\}$

$= \{\frac{1}{2}, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \dots\}$

Lagrange's Theorem

- **Lagrange theorem** is one of the central theorems of abstract algebra. It states that in group theory, for any finite group say G , the order of subgroup H of group G divides the order of G . The order of the group represents the number of elements.

Group Homomorphism

Homomorphism:

Let $(G, *)$ and $(G', \circ)$ two groups.

$$f: G \rightarrow G'$$

A mapping f from $(G, *)$ to $(G', \circ)$ is called group homomorphism if

$$f(x * y) = f(x) \circ f(y)$$

$f(x)$ $f : (R^*, +) \rightarrow (R^*, \cdot) \quad [R^* = R - \{0\}]$

$$f(x) = e^x$$

Here,

$$f(x+y) = e^{x+y}$$
$$= e^x \cdot e^y$$

$$= f(x) \cdot f(y)$$

So, there is homomorphism.

Group Isomorphism

Isomorphism :

For two groups $(G, +)$ and $(G', *)$ a mapping $f: G \rightarrow G'$ is called isomorphism if

- f is one-one
- f is onto
- f is homomorphism i.e. $f(a + b) = f(a) * f(b) \forall a, b \in G$.

Let G be additive Group of Integers.

Let G be " " " even integers.

$$f: G \rightarrow G'$$

$$f(x) = 2x$$

1. Check for one to one

$$\text{Let, } f(x) = f(y)$$

$$\Rightarrow 2x = 2y$$

$$\Rightarrow x = y$$

So, it is one to one

2. Check for onto

$$\text{Let, } y = 2x$$

$$x = \frac{y}{2}$$

$$\text{So, } f(x) = 2 \times \frac{y}{2} = y \quad \text{So, it is}$$

also onto

3. Check for homomorphism

$$f(x+y) = f(x) + f(y)$$

$$2(x+y)$$

$$= 2x + 2y = f(x) + f(y)$$

Group Automorphism

- A group automorphism is an isomorphism from a group to itself.

Topics in Syllabus

- Algebraic Structure
- Semi Group
- Monoid
- Group
- Abelian Group
- Subgroup
- Addition and Multiplication modulo
- Order of element
- Generating element and cyclic group
- Coset and Lagrange's theorem
- Homomorphism, Isomorphism, Automorphism
- Normal Subgroup
- Basic Definitions : Ring, Integral Domain, Field.